O. Priyanjali
23011102058
IoT-A

Caesar Cipher

Encryption Algorithm

1. Start
2. Input plaintext (text)
3. Input key (key)
4. Initialize

   hash-value = 0

   choose prime p = 31

   choose large modulus $m = 10^9 + 7$
5. For each character ch in key:

   hash-value = (hash-value × p + ASCII (ch)) mod m
6. compute shift

   shift = hash-value mod 26
7. For each character in plaintext:

   If uppercase:

   C = (P - 'A' + shift) mod 26 + 'A'

   If lowercase:

   C = (P - 'a' + shift) mod 26 + 'a'

   Else:

   keep character unchanged
8. Output encrypted text
9. Stop

Decryption Algorithm

1. start
2. input ciphertext (text)
3. Input key (key)
4. compute same hash as encryption
5. For each character in ciphertext:

   If uppercase:

   $p = (c - 'A' - shift + 26) \bmod 26 + 'A'$

   If lowercase:

   $p = (c - 'a' - shift + 26) \bmod 26 + 'a'$

   Else:

   keep unchanged
6. output decrypted text
7. stop

Hash function

hash = some big number → then reduced to shift

→ we convert a key into a number

→ polynomial rolling hash is used

hash = (hash × p + ASCII) mod m

working of polynomial hash

key = "abc"

p = 31

hash = 0

for 'a':

hash = (0 × 31 + 97) = 97

For 'b':

hash = (97 x 31 + 98) = 3105

For 'c':

hash = (3105 x 31 + 99) = 96354

Final hash = 96354 (then mod m)

Example Implementation

Plaintext = hello

key = abc

p = 31

$m = 10^9 + 7$

→ hash = (hash x p + ASCII) mod m

start: hash = 0

For 'a' → ASCII = 97

hash = (0 x 31 + 97) = 97

For 'b' → ASCII = 98

hash = (97 x 31 + 98) = 3105

For 'c' → ASCII = 99

hash = (3105 x 31 + 99) = 96354

hash = 96354

→ shift = hash % 26

= 96354 % 26 = 8

Encryption

c = (p + shift) % 26

h → (7+8) % 26 = 15 p

e → (4+8) % 26 = 12 m

l → (11+8) % 26 = 19 t

o → (14+8) % 26 = 22 w

Decryption

$P = (c - shift + 26) \% 26$

$p \rightarrow (15-8) \% 26 = 7$ h

$m \rightarrow (12-8) \% 26 = 4$ e

$t \rightarrow (19-8) \% 26 = 11$ l

$t \rightarrow (19-8) \% 26 = 11$ l

$w \rightarrow (22-8) \% 26 = 14$ o

hash value = 96354

shift = 8

encrypted text = pmttw

decrypted text = hello